

Group 1

Master Control Predictive Modeling

03

IS6813

2026

Agenda

01 Business Problem

02 Preprocessing Steps

03 Contact Lead Job Title – Cleaning
and Analysis

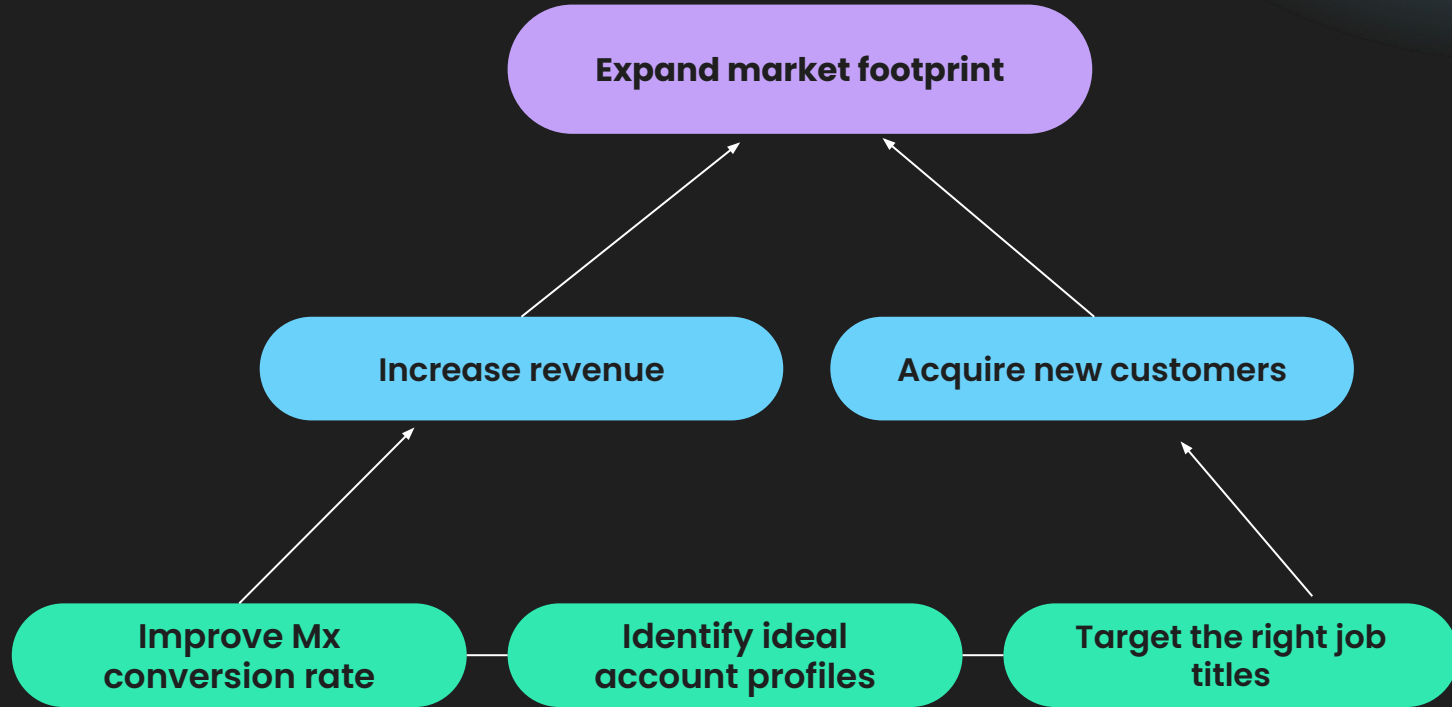
04 Exploratory Analysis

05 EBM Model

06 Hypothetical Scenarios

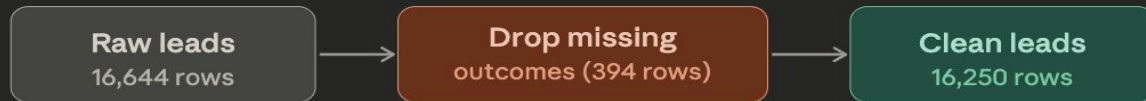
07 Business Impact

Business Problem

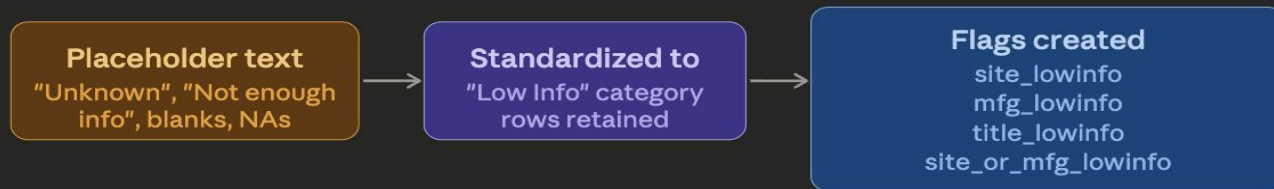


Preprocessing

Missing values



Low information flag

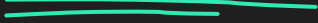


Why it matters — Mx conversion rate by enrichment



Missing data is not noise — it is a predictive signal retained in all models

Contact Lead Job Title – Cleaning & Analysis



Pre Processing Steps

Remove

- Stop words
- Letter case
- Numbers
- Punctuation
- Special Symbols

Common Word Map

Common word conversions

- Typos
- Abbreviations
- Short forms

Unique Title Reductions

Unique titles from 5,791
down to 5,244

Create Single Word Tokens

Minimum 5 occurrences
retained

361 titles

Top Job Title Word – Conversion

1 Production	2 Officer	3 Project	4 Research	5 Chief
24.0% - Conversion Rate	23.8%	23.2%	23.1%	21.7%
2.5% - Occurrence Rate	4.1%	2.8%	1.3%	3.6%
6 Assistant	7 Lead	8 Compliance	9 Systems	10 President
20.6%	20.0%	19.4%	18.8%	18.5%
1.2%	2.7%	2.4%	2.2%	4.9%

MX Conversion by Acct Manufacturing Model

Baseline MX conversion rate: 12.9%

Acct Manufacturing Model	Leads	% of MX	Conv. Rate (%)	Lift
Original Equipment Manufacturing	54	1.31	50.00	3.87
CMO	249	6.04	24.90	1.93
CDMO	415	10.06	17.83	1.38
In-House	1,486	36.02	17.29	1.34
Services / Consulting	113	2.74	16.81	1.30
Distribution / Wholesale	43	1.04	13.95	1.08
Research Institute	31	0.75	9.68	0.75
CRO	65	1.58	9.23	0.71
Low Info	1,530	37.09	2.22	0.17

MX Conversion by Acct Primary Site Function				
Baseline MX conversion rate: 12.9%				
Acct Primary Site Function	Leads	% of MX	Conv. Rate (%)	Lift
Specialty Chemical Manufacturing	100	2.42	37.00	2.86
Nutraceutical manufacturing	71	1.72	35.21	2.72
Radiopharmaceuticals	35	0.85	34.29	2.65
Cosmetics manufacturing	30	0.73	33.33	2.58
Single-use disposables	32	0.78	31.25	2.42
Lab Instruments	74	1.79	24.32	1.88
Cell therapies (CAR-T, NK)	98	2.38	21.43	1.66
Gene therapies	31	0.75	19.35	1.50
In-vitro diagnostic (IVD) kits	88	2.13	19.32	1.50
Drug discovery & development	58	1.41	17.24	1.33
Surgical instruments	59	1.43	16.95	1.31
Implantable devices	130	3.15	16.92	1.31
Sterile injectables	144	3.49	16.67	1.29

MX Conversion by Last Tactic Campaign Channel

Baseline MX conversion rate: 12.9%

Last Tactic Campaign Channel	Leads	% of MX	Conv. Rate (%)	Lift
SEO	452	10.96	26.99	2.09
Direct/Inbound	333	8.07	24.32	1.88
Events	187	4.53	21.93	1.70
Directory Listing	104	2.52	19.23	1.49
Online Ads	784	19.01	18.75	1.45
Outbound Prospecting	111	2.69	15.32	1.19
Low Info	37	0.90	13.51	1.05
Email	1,142	27.68	6.04	0.47
External Demand Gen	932	22.59	2.90	0.22

Finding Mx's Best Leads

What a Predictive Model Reveals About Who Actually Converts

Explainable Boosting Machine (EBM) Results — Josh

The Model That Shows Its Work

Traditional Black-Box Model

- High accuracy
- Score arrives with no per-lead explanation
- Can't tell sales which attributes drove the result
- Hard to audit

Explainable Boosting Machine (EBM)

- 0.86 AUC — best accuracy tested
- Prediction = sum of each feature's effect
- Each feature gets its own shape function
- Pairwise interactions detected automatically
- Every score is fully decomposable

Shape functions mean a sales rep can ask 'why did this lead score high?' — and get a precise, data-backed answer for every attribute.

01. The model learns a lookup function for each feature independently.

02. Boosting iterates through these curves, correcting errors, until they stabilize.

03. At prediction time, each feature's curve value is added together.

What Predicts Conversion May Surprise You

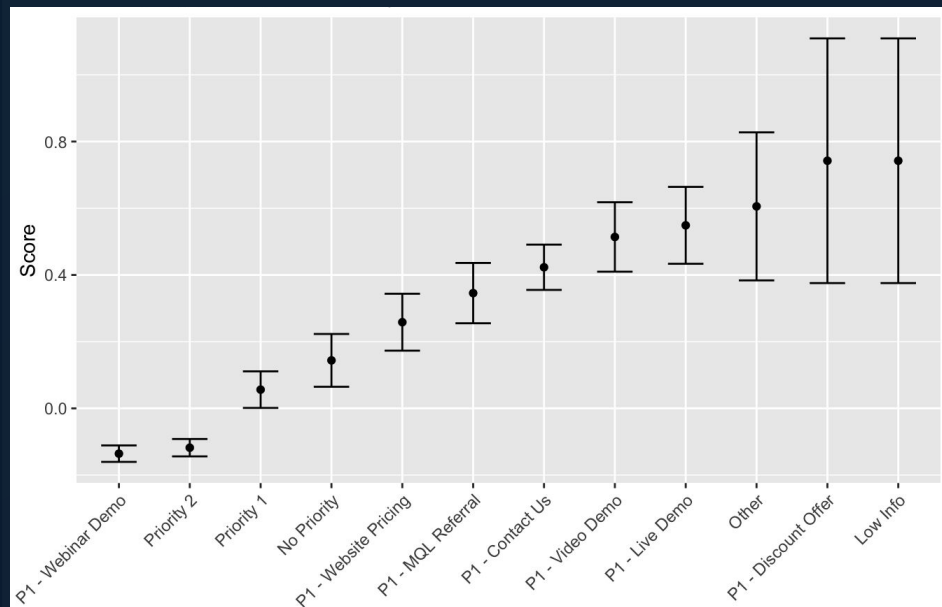
Marketing teams typically segment on industry and territory. The model says those matter far less than how a lead engaged.

What the model says drives conversion

Engagement type (how they reached out), Job title (who they are), Campaign channel (how they found us)

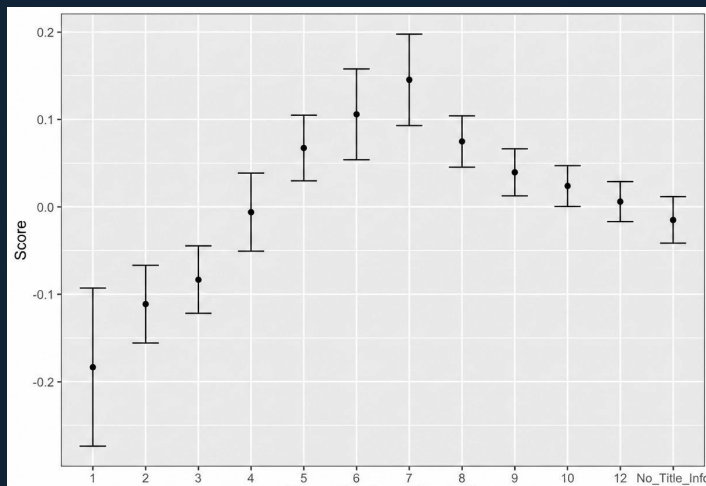
What the model says matters less than expected

- Industry — minimal effect once engagement is controlled for.
- Territory — only Americas is meaningfully positive.
- Account size — modest signal at best.



Rightmost categories (Other, Discount Offer, Low Info) show high scores but wide confidence intervals due to small sample sizes — estimates are unreliable

Job Title Tells You Who to Call — and Who to Reroute



Who the lead is matters more

Title cluster effect on conversion is second only to engagement type - larger than industry, territory or account size

Cluster 7 — Prioritize for MX

- Founders, Production, Plant, and Engineering Managers
- These roles feel manufacturing quality pain directly
- Strongest positive conversion signal (+0.15)
→ **First call for MX outreach**

Cluster 1 — Route to QX

- QA Directors, Quality Specialists, Associate Directors of Quality
- Lowest-scoring group for MX conversion (−0.19)
- Strong potential fit for MasterControl's QX product
→ **Don't discard — redirect**

The next section shows exactly what a Cluster 7 and a Cluster 1 lead look like as real prospects.

Ideal Customer Profile

Attribute	Target Profile	Deprioritize
→ Industry	→ Medical Device, Blood & Biologics	→ Non-Life Science, Unknown
→ Priority	→ P1 - Contact Us, P1-Video/Live Demo	→ P1 - Webinar Demo, Priority 2
→ Territory	→ Americas	→ APAC & Oceania, Japan
→ Account Tier	→ Small	→ Large
→ Channel	→ Direct/Inbound, Events	→ Email, External Demand Gen
→ Job Title	→ Founders, Production/Plant/Engineering Managers	→ QA Directors, Quality Specialists

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Hypothetical Scenario

Sarah



Industry	Medical Device
Job Title	Production Manager
Channel	SEO
Territory	Americas
Priority	Priority 1
Manufacturing Type	In-House

David



Industry	Unknown
Job Title	Quality Assurance Director
Channel	Online Ads Channel
Territory	APAC & Oceania
Priority	No Priority
Manufacturing Type	Services/Consulting

Hypothetical Scenario



PERFORMANCE PROBABILITIES

→ SUCCESSFUL = .972

→ NOT SUCCESSFUL = .028

→ LIFT OVER BASELINE 7.52X

Industry	Medical Device
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Job Title	Production Manager
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Channel	SEO
---------	-----

Territory	Americas
-----------	----------

Priority	Priority 1
----------	------------

Manufacturing Type	In-House
--------------------	----------

Hypothetical Scenario

David



PERFORMANCE PROBABILITIES

→ SUCCESSFUL = .025

→ NOT SUCCESSFUL = .975

→ LIFT OVER BASELINE .19X

Industry	Unknown
Job Title	Quality Assurance Director
Channel	Online Ads Channel
Territory	APAC & Oceania
Priority	No Priority
Manufacturing Type	Services/Consulting

Business Impact



Key Values

Parameter	Value	Source
1) Average Mx product revenue - Qx proxy 2) Estimated cost per QAL outreach 3) Probability Threshold	1) \$70,000 2) \$100-\$1000 3) .1-.7	1) Provided (Qx product cost) 2) Sales rep time + campaign cost 3) Using EBM Predictions

Best Threshold By Cost Level

cost_per_outreach <dbl>	p_success_threshold <dbl>	pursued_per_1000 <dbl>	conversions_per_1000 <dbl>	no_conversion_per_1000 <dbl>	expected_revenue_1000 <dbl>	expected_cost_1000 <dbl>	net_value_1000 <dbl>
100	0.1	300.1	104.7	195.4	7326247	30008	7296239
200	0.1	300.1	104.7	195.4	7326247	60016	7266231
300	0.1	300.1	104.7	195.4	7326247	90025	7236222
400	0.1	300.1	104.7	195.4	7326247	120033	7206214
500	0.1	300.1	104.7	195.4	7326247	150041	7176206
600	0.1	300.1	104.7	195.4	7326247	180049	7146198
700	0.1	300.1	104.7	195.4	7326247	210057	7116190
800	0.1	300.1	104.7	195.4	7326247	240065	7086182
900	0.1	300.1	104.7	195.4	7326247	270074	7056173
1000	0.1	300.1	104.7	195.4	7326247	300082	7026165

Best Threshold By Cost Level

cost_per_outreach <dbl>	p_success_threshold <dbl>	pursued_per_1000 <dbl>	conversions_per_1000 <dbl>	no_conversion_per_1000 <dbl>	expected_revenue_1000 <dbl>	expected_cost_1000 <dbl>	net_value_1000 <dbl>
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Wrap Up



Model Performance

ROC = .86

- **How a lead engages matters more than what company they come from**
 - Strongest predictor → Priority signal.
 - Effect on conversion is ~3-5x larger than industry, territory, or account size.
 - Targeting should shift from firmographic towards behavioral signals.
- **Quality-titled leads aren't Mx leads – they're Qx leads.**
 - Strongest negative signal → QA Directors and Quality Specialists
 - Strongest positive signal → Production and Plant Managers
 - **Recommendation:** not drop the quality-titled leads – reroute them to Qx, turning Mx loss to a Qx opportunity.

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Thank you!